

Orthogonal Waveform Design for Integrated Sensing and Communication Networked Base Stations in Low-Altitude Sensing Scenarios

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Abstract—This paper proposes an orthogonal waveform design method for integrated sensing and communication (ISAC) networked base stations in low-altitude sensing (LAS) scenarios. Considering the large number of base stations, they are first grouped by carrier frequencies, and orthogonal waveforms are designed for intra-group base stations to suppress mutual interference. Given the limited available spectrum of base station, the polyphase-coded frequency modulation (PCFM) waveforms are optimized so that the final orthogonal waveforms can concentrate the energy in the expected frequency band. Therefore, an optimization model based on the phase differences of PCFM waveforms is formulated. The objective of this model is to minimize both the peak cross-correlation level (PCCL) and the auto-correlation peak sidelobe level (APSL). This model is solved efficiently by the sequential quadratic programming (SQP) algorithm. Simulation results confirm that the proposed orthogonal waveforms exhibit low APSL and PCCL. Their spectral energy is concentrated, maintaining the characteristics of the PCFM waveform.

Index Terms—LAS, networked base stations, orthogonal waveform design, PCFM, SQP.

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I. INTRODUCTION

With the rapid development of unmanned aerial vehicle (UAV) technologies, the number of UAVs in urban low-altitude airspace has been increasing significantly [1], [2], [3]. UAVs have demonstrated great potential in logistics distribution, aerial photography, and emergency response. However, their widespread adoption also brings new challenges to public safety, airspace management, and personal privacy protection [4]. Therefore, establishing effective mechanisms for UAV management has become a critical component of low-altitude safety systems. In this context, reliable low-altitude sensing (LAS) is essential for detecting and tracking UAVs.

In urban areas, a large number of base stations have been deployed. Besides, the hardware of base stations can support the implementation of integrated sensing and communication (ISAC) systems [5]. Therefore, utilizing existing base stations for LAS is a cost-effective and implementable solution. While a single base station often suffers from limited sensing accuracy, blind spots, and easily signal blockages, networked base stations can effectively compensate for these limitations. Such networked base stations provide broader spatial coverage and enhanced sensing reliability and have been widely studied. In work [6], Liu et al. proposed a symbol-level fusion technique and a three-dimensional grid-based discrete Fourier transform algorithm to improve positioning performance in heterogeneous base station environments. The work [7], [8]

focused on optimizing beamforming strategies for dual-base-station ISAC architectures and coordinated multi-point joint transmission (CoMP-JT) in multi-static systems, using the successive convex approximation (SCA) algorithm. Zhao et al. [3] presented a cooperative model involving multiple base stations tailored to UAV scenarios, while Li et al. [9] introduced a hybrid sensing architecture combining single-station three-dimensional sensing with a cellular multi-station topology to support seamless LAS. Nevertheless, none of these studies have considered inter-base-station interference from the signal processing view.

In practical scenarios, due to the large number of base stations, this study first employs frequency division to divided the base stations into multiple groups. To mitigate interference among intra-group base stations, we draw inspiration from the concept of orthogonal waveforms in multiple-input multiple-output (MIMO) radar systems to design waveforms. However, the spectral energy of conventional orthogonal waveforms is relatively dispersed [10]. Meanwhile, the designed waveforms are dedicated exclusively to sensing, while the communication and sensing functionalities of the base station are separated in the time domain via time-division multiplexing (TDM) [11]. Considering the limited available spectrum of base station, it is critical to concentrate waveform energy within the expected frequency band. To address this challenge, we adopt polyphase-coded frequency modulation (PCFM) waveforms [12], [13], [14] which are known for their constant-modulus structure, smooth phase continuity, and highly concentrated spectral energy characteristics.

Based on the aforementioned analysis, this paper proposes an orthogonal waveform design method for ISAC networked base stations in LAS scenarios. The primary contributions of this work can be summarized as follows:

- An orthogonal waveform optimization model is formulated based on the phase differences of PCFM waveforms. The optimized waveforms achieve low auto-correlation peak sidelobe level (APSL) and peak cross-correlation level (PCCL). Their spectral energy is concentrated, maintaining the characteristics of the PCFM waveform.
- The proposed orthogonal waveform design is validated under a carrier-frequency-based grouping strategy of networked base stations. Simulation results confirm that the resulting waveforms maintain highly concentrated spectral energy and effectively reduce inter-base-station interference.

II. PCFM WAVEFORM

PCFM is a type of waveform characterized by continuous phase, constant modulus, and well-controlled spectral properties. Originating from the continuous phase modulation (CPM) technique commonly used in communication systems, PCFM is particularly well suited for applications involving polyphase radar coding [12]. Its implementation is depicted in Fig.1. Here, α_N represents the phase variation determined by the

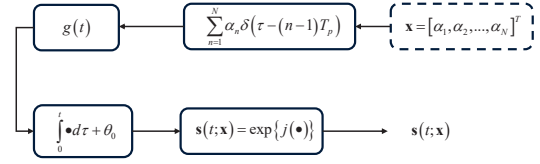


Fig. 1: Implementation of PCFM waveforms.

successive chips of the polyphase code, and its expression is given

$$\alpha_n = \Psi(\tilde{\alpha}_n) = \begin{cases} \tilde{\alpha}_n & |\tilde{\alpha}_n| \leq \pi \\ \tilde{\alpha}_n - 2\pi \text{sgn}(\tilde{\alpha}_n) & |\tilde{\alpha}_n| > \pi \end{cases} \quad (1)$$

as follows

$$\tilde{\alpha}_n = \theta_n - \theta_{n-1}, n = 1, 2, \dots, N \quad (2)$$

where $\text{sgn}(\bullet)$ is the sign function and θ_n is the phase value of the n -th chip in the length $N + 1$ polyphase code.

In Fig.1, the implementation of the PCFM waveform begins with $\mathbf{x} = [\alpha_1, \alpha_2, \dots, \alpha_N]^T$, which generates an impulse train of N with a uniform time separation of T_p . Each impulse in the sequence is weighted by a corresponding coefficient α_N . This weighted impulse train is passed through a shaping filter $g(t)$ producing a continuous-phase signal as the output.

Consequently, the PCFM waveform can be expressed as

$$\mathbf{s}(t; \mathbf{x}) = \exp \left\{ j \left(\int_0^t g(\tau) * \left[\sum_{n=1}^N \alpha_n \delta(\tau - (n-1)T_p) \right] d\tau + \theta_0 \right) \right\} \quad (3)$$

The $g(t)$ is required to have a normalized integral, and its duration is constrained to $[0, T_p]$. Common choices for shaping filters include the rectangular filter and the raised-cosine filter. The code length L of the PCFM waveform is defined as

$$L = KN \quad (4)$$

where K is the number of samples per chip.

III. PROPOSED METHOD

Consider a set of orthogonal waveforms with M base stations and a code length of L . The corresponding waveforms can be derived from (3)

$$\mathbf{s} = [\mathbf{s}_1(t; \mathbf{x}), \mathbf{s}_2(t; \mathbf{x}), \dots, \mathbf{s}_M(t; \mathbf{x})] \quad (5)$$

as follows

$$\mathbf{s}_m(t; \mathbf{x}) = \exp(j\Omega_m(t)), m = 1, 2, \dots, M \quad (6)$$

$$\Omega_m(t) = j \left(\int_0^t g(\tau) * \left[\sum_{n=1}^N \alpha_{mn} \delta(\tau - (n-1)T_p) \right] d\tau + \theta_m \right), n = 1, 2, \dots, N, m = 1, 2, \dots, M \quad (7)$$

where α_{mn} is the n -th phase difference of the m -th base station and θ_m is the initial phase of the m -th base station.

The mathematical expression for the rectangular filter is given by

$$g(t) = \begin{cases} \frac{1}{T_p} & 0 \leq t \leq T_p \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

Based on (6) and the definitions of related functions, the aperiodic auto-correlation function of signal $\mathbf{s}_m$ is expressed as

$$A(\mathbf{s}_m, k) = \begin{cases} \frac{1}{L} \sum_{n=1}^{L-k} \mathbf{s}_m(n) \mathbf{s}_m^*(n+k), & 0 \leq k < L \\ \frac{1}{L} \sum_{n=-k+1}^L \mathbf{s}_m(n) \mathbf{s}_m^*(n+k), & -L < k < 0 \end{cases} \quad (9)$$

$, m = 1, 2, \dots, M$

where $\mathbf{s}_m$ is a shorthand for $\mathbf{s}_m(t; \mathbf{x})$, and this shorthand will be used in subsequent formulas. The aperiodic cross-correlation function between $\mathbf{s}_m$ and $\mathbf{s}_{\tilde{m}}$ is expressed as

$$C(\mathbf{s}_m, \mathbf{s}_{\tilde{m}}, k) = \begin{cases} \frac{1}{L} \sum_{n=1}^{L-k} \mathbf{s}_m(n) \mathbf{s}_{\tilde{m}}^*(n+k), & 0 \leq k < L \\ \frac{1}{L} \sum_{n=-k+1}^L \mathbf{s}_m(n) \mathbf{s}_{\tilde{m}}^*(n+k), & -L < k < 0 \end{cases} \quad (10)$$

$, m \neq \tilde{m}, m, \tilde{m} = 1, 2, \dots, M$

Therefore, the APSL of the orthogonal waveform is expressed as

$$I_{APSL} = \max_{m=1,2,\dots,M} \left\{ \max_{k \neq 0} |A(\mathbf{s}_m, k)| \right\} \quad (11)$$

The PCCL is expressed as

$$I_{PCCL} = \max_{m \neq \tilde{m}} \left\{ \max_{k=-L+1, \dots, L-1} |C(\mathbf{s}_m, \mathbf{s}_{\tilde{m}}, k)| \right\} \quad (12)$$

$, m, \tilde{m} = 1, 2, \dots, M$

The APSL reflects the pulse compression capability of a waveform, and the PCCL measures the extent of mutual interference between different waveforms.

The core challenge in orthogonal waveform design is to optimize the phase differences such that both APSL and PCCL are minimized simultaneously. It is expressed as

$$\begin{aligned} \min_{\alpha, \varepsilon} \quad & \varepsilon \\ \text{s.t.} \quad & |A(\mathbf{s}_m, k)| \leq \varepsilon, m = 1, 2, \dots, M, k = 1, 2, \dots, L \\ & |C(\mathbf{s}_m, \mathbf{s}_{\tilde{m}}, k)| \leq \varepsilon, m \neq \tilde{m}, m, \tilde{m} = 1, 2, \dots, M, \\ & k = 1, 2, \dots, L \end{aligned} \quad (13)$$

where α is a column vector comprising MN variables. It is a vectorization that corresponds to all phase differences of the orthogonal waveform $\mathbf{s}$.

This optimization model is formulated based on a peak sidelobe level (PSL) minimization criterion. In contrast to alternatives such as integrated sidelobe level (ISL) minimization, this criterion places a direct bound on the worst-case sidelobe.

This characteristic is particularly beneficial in practice, as it reduces the likelihood of false alarms and mitigates the risk that weak targets are masked by elevated sidelobes.

The physical interpretation of this optimization model is that it constrains all auto-correlation sidelobes and cross-correlation values to remain below an upper threshold. This formulation is a nonlinear programming problem involving $MN + 1$ variables and $M(3N - 2)$ nonlinear constraints, all of which are twice continuously differentiable. The resulting problem can be efficiently solved by the sequential quadratic programming (SQP) algorithm. The proposed optimization model is implemented using the `fmincon` function from MATLAB's Optimization Toolbox, with the SQP algorithm selected as the solver. Without prior information, θ is assumed to be uniformly distributed over $[0, 2\pi]$. The detailed procedure is described in **Algorithm 1**.

Algorithm 1 Proposed PCFM Orthogonal Waveform Algorithm

Require: Initial phase values $\theta(n), n = 1, 2, \dots, M(N + 1)$.

Ensure: Orthogonal waveforms $\mathbf{s}_m(t; \mathbf{x}), m = 1, 2, \dots, M$.

- 1: Calculate the phase variations $\mathbf{x}_m = [\alpha_{m1}, \alpha_{m2}, \dots, \alpha_{mN}]^T, m = 1, 2, \dots, M$ according to (1) and (2).
 - 2: The original variable $\mathbf{x}_m = [\alpha_{m1}, \alpha_{m2}, \dots, \alpha_{mN}]^T, m = 1, 2, \dots, M$ is consolidated into the column vector α .
 - 3: Input α into the optimization model given by (13).
 - 4: Invoke the SQP algorithm to solve the model.
 - 5: Obtain the optimal phase variations α , and derive the final orthogonal waveforms based on (2) and (6).
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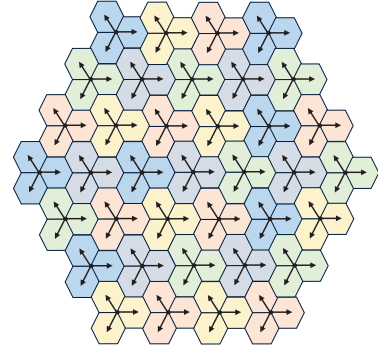


Fig. 2: Layout diagram of base stations with five-frequency-division.

IV. SIMULATION RESULTS

This section evaluates the effectiveness of the orthogonal waveform design through a series of simulation experiments. The pulse width is $4 \mu s$, the sampling frequency is 120 MHz, and the code length is $L = 480$. The carrier frequency is 4.9 GHz, and its available frequency band is 200 MHz. Unless specially stated otherwise, the above statements serve as the default configuration for all subsequent simulations.

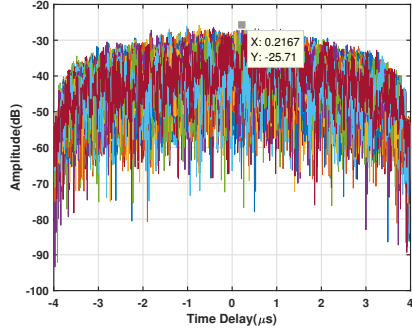


Fig. 3: Simulation results for inter-group base stations (with an available bandwidth of 40 MHz).

Considering the large number of base stations in urban environments, the base stations are grouped by carrier frequencies. The orthogonal waveform proposed in this study is applied to intra-group base stations to suppress mutual interference between base stations. In Fig.2, the simulation scenario comprises 37 base stations. All the base stations are divided into 5 groups. Base stations with the same color belong to the same group. The number of base stations in each group is 7, 7, 7, 8, and 8 respectively.

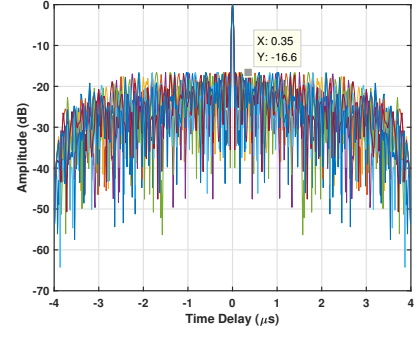
A. Inter-group waveforms

Inter-group waveforms denote waveforms from different groups with different carrier frequencies. Following the above scheme for grouping network base stations, the available bandwidth for each group is designated as 40 MHz. As demonstrated in the following subsection, the vast majority of waveform energy for each group is concentrated within the frequency band of [-15 MHz, 15 MHz]. Accordingly, a 10 MHz guard band is introduced between adjacent groups to prevent potential inter-group interference caused by spectral leakage. Assuming that waveforms corresponding to 7 and 8 base stations are respectively modulated at carrier frequencies of 4.9 GHz and 4.94 GHz. The resulting cross-correlation function is illustrated in Fig.3, where the PCCL is -25.71 dB. This result indicates that the designed waveforms effectively suppress inter-base-station interference.

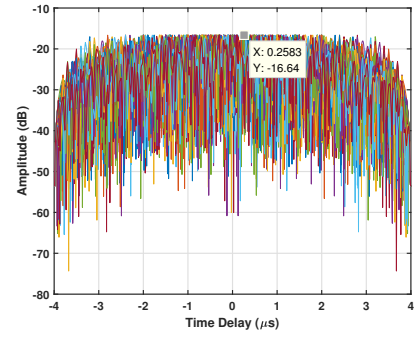
B. Intra-group waveforms

Due to the application of carrier frequency division, the auto-correlation and cross-correlation outcomes for each intra-group waveforms are similar. By adjusting the number of chips N and the number of sampling points per chip K in the PCFM waveform, the designed waveforms energy is maximally concentrated within the 30 MHz bandwidth [12]. To fully demonstrate the efficacy of the proposed algorithm, a group with $M = 8$ is selected for simulation validation. Fig.4 present the auto-correlation level, cross-correlation level, and frequency-domain energy distributions for intra-group base station configurations of $M = 8$. It can be seen that auto-correlation and cross-correlation levels remain below -16 dB. Additionally, the vast majority of waveform energy

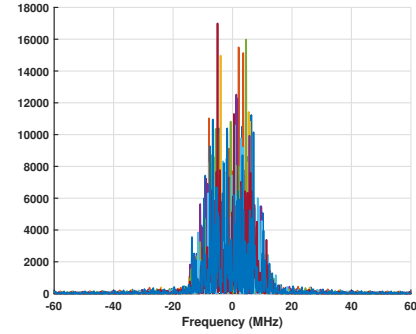
is concentrated within the frequency band of [-15 MHz, 15 MHz] .



(a)



(b)



(c)

Fig. 4: Simulation results for intra-group base station number $M = 8$: (a) The auto-correlation level. (b) The cross-correlation level. (c) Frequency-domain energy distribution.

V. CONCLUSION

This study presents an orthogonal waveform design method for ISAC networked base stations in LAS scenarios. The proposed waveforms achieve both low APSL and PCCL, while maintaining the concentrated spectral energy characteristics inherent to PCFM waveforms. Moreover, when base stations are grouped by carrier frequency, the designed waveforms effectively suppress inter-base-station interference, demonstrating the method of applicability in ISAC networked base stations.

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